An Objective Eye Exam from a Consumer EEG Headband and a Consumer VR Headset

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***Abstract*—We turn the eye-as-a-camera rig, a Muse headband with one electrode added over the visual cortex and a Meta Quest 3S running a stimulus page in its browser, into an eye test that asks nothing of the subject but to look at a dot with one eye covered. Four tests run as ON/OFF blocks of flicker: a grating appearing and disappearing on a grey of the same mean luminance with its bars shrinking (acuity) or its contrast falling (contrast), one sector of a 4×4 grid flickering at a time (field), and the patch in red, green and blue (colour). Each step is scored by how much of the flicker frequency the EEG carries against the test's OFF blocks, with an exact permutation test; acuity and contrast thresholds follow the sweep-VEP rule, a line through the rising limb of amplitude against log bar width or log contrast extrapolated to zero. A hardware-free phantom with set cut-offs, a dead sector and colour gains passes all twelve checks of a full-stack gate. The first sitting on a person fitted no threshold, because both ear electrodes sat at 100 to 280 μV under the headset strap and the signal check had passed them; it produced the contact check that now runs before each eye. Nothing here is a diagnosis: it is the apparatus and its validation for a report-free visual screen on hardware a health worker can carry.**

***Index Terms*—SSVEP; EEG; visual acuity; sweep VEP; extended reality; consumer electronics; mobile health; visual screening.**

I. Introduction

Mann et al. showed that the steady-state visual evoked potential (SSVEP) read from a few scalp electrodes can turn the eye into a camera: a square flickers at one position after another and the response at the flicker frequency says how bright each position was [1]. We built that rig on consumer hardware, a Muse headband with one cup electrode added at Oz and a Meta Quest 3S whose browser renders the stimulus (Fig. 2a), and read words, colours and a sealed optotype from one subject's EEG. The same measurement underlies the sweep visual evoked potential, in which the response to a grating is followed as the grating gets finer and the acuity is the bar width at which it is extrapolated to vanish [2], and the electrophysiological contrast threshold, the contrast at which the response to a grating of fixed size vanishes [3]. Both are objective: the subject reports nothing and need not read, which is what a visual screen for community health work needs. This paper describes the eye exam built on the rig, its validation without hardware, and what its first sitting on a person taught us.

II. Method

The subject looks at a central dot with one eye covered while the page runs each test as ON/OFF blocks (4 s each, two pairs per step) at one flicker frequency, frame-exact on the

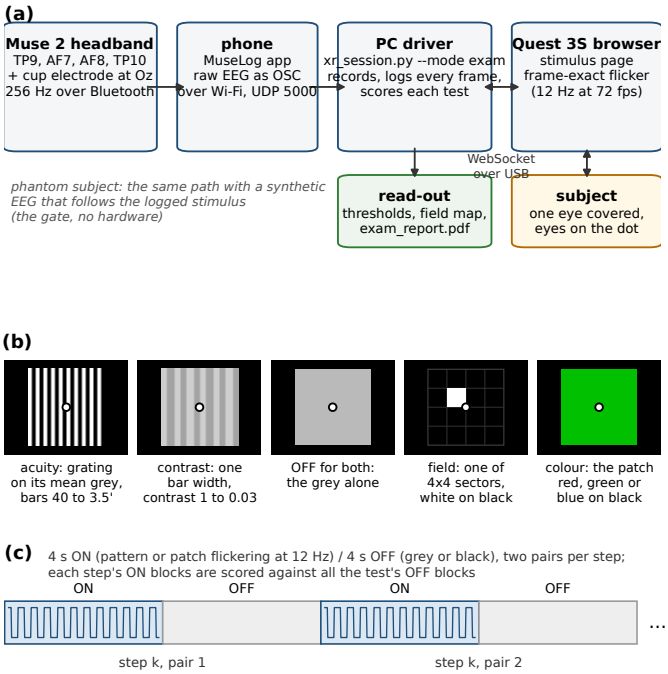


Fig. 1. The exam. (a) The path from scalp to read-out: the headband streams raw EEG through the phone to the PC, which drives the stimulus page in the headset's browser over USB, logs every frame it draws, and scores each test; the phantom subject replaces the first two boxes with a synthetic EEG that follows the logged stimulus. (b) What the headset shows in the four tests, and the OFF screen of the grating tests. (c) The block timeline of one step.

headset's 72 Hz display (12 Hz, three frames on and three off); Fig. 1 shows the path, the four stimuli and the timeline, and Fig. 2 the page as rendered. For acuity a vertical black/white grating appears and disappears on a grey of the same mean luminance, so only the pattern comes and goes, its bar width stepping from 40 to 3.5 arcmin; for contrast the grating keeps one bar width while its Michelson contrast steps from 1 to 0.03; for the field one sector of a 4×4 grid flickers white against black while the eyes stay on the dot; for colour the whole patch flickers red, green or blue. Bar widths are set in arcmin from the display's pixels per degree, rounded to whole device pixels, and the width delivered is reported with every block. Every frame and block edge is logged, and each step is scored on the channel with the largest ON/OFF difference over the test: line power at the flicker frequency and its double against the flanking spectrum as a log signal-to-noise ratio per block, the step's ON blocks against all the test's OFF blocks by an exact permutation test over the block labels, a

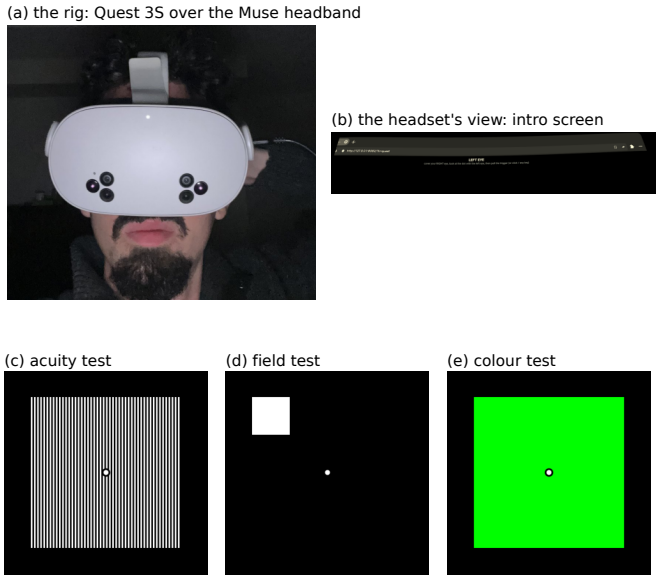


Fig. 2. The rig and the page. (a) The Quest 3S worn over the Muse headband (the cup electrode at Oz is under the strap). (b) The headset's own view of the intro screen of one eye, captured from the device. (c)–(e) The stimulus page during an ON frame of the acuity, field and colour tests, captured from the browser: the grating on its own grey, one white sector of the 4×4 grid, the green plane, each with the fixation dot.

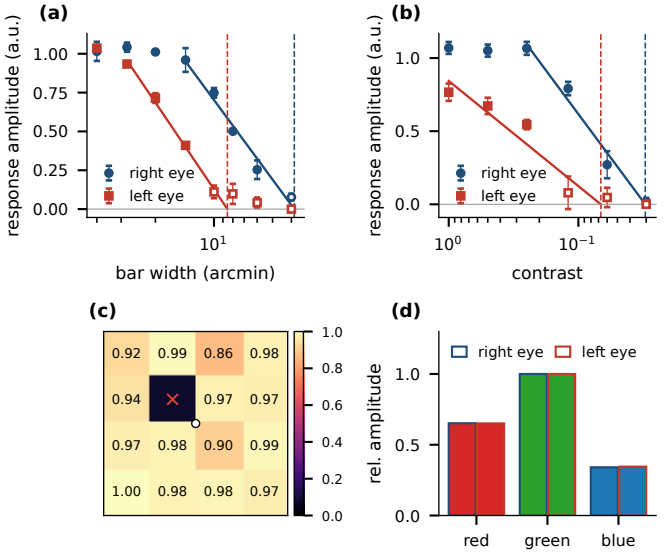


Fig. 3. The phantom gate's read-out. (a) Acuity and (b) contrast: noise-corrected amplitude at the flicker frequency and its double per bar width and per contrast for the right (circles) and left (squares) phantom eye; open markers are not significant ($p \geq 0.05$); lines are the fitted limbs, dashed verticals their zero crossings. (c) The right eye's field: amplitude per sector relative to the best, the cross the sector the phantom cannot see, the dot the fixation point. (d) Colour planes of both eyes relative to the best plane.

noise-corrected amplitude, and the response phase against the logged flicker edges (relative between sectors and eyes, not a latency). A channel whose rms stays above 60 μV through a test is set aside as out of contact. Thresholds follow the sweep-VEP rule [2]: from the step just below the finest step that still responds, through it and up the rising limb, a line is fitted and extrapolated to zero amplitude; a threshold is refused when only one step responds, and one below the range shown is said so rather than turned into a Snellen value. The field is reported as amplitude and phase per sector with the silent sectors named and, with two eyes, their sector-by-sector ratio and phase difference; colour as the amplitude of each plane.

III. Validation without hardware

The rig is gated against a phantom subject that reads the stimulus log as it is written and streams synthetic EEG whose SSVEP follows what is on screen at the frequency actually delivered. For the exam the phantom's eye got a logistic visibility in log bar width and log contrast with a cut-off per eye, sectors that give no response, and a gain per colour plane. The gate dials in a right eye with cut-offs of 8 arcmin and contrast 0.08 and one dead sector, a left eye twice as coarse and whole, and gains of 0.7, 1 and 0.4 for red, green and blue, runs the whole stack (driver, headless browser, phantom; 3 s blocks, 13.7 minutes) and asks twelve things: every stage delivered, each fitted threshold inside the band where the phantom's visibility runs from 0.03 to 0.6 (a line through the rising limb crosses zero below the half-visibility point), the left eye at least $1.3\times$ coarser on both thresholds, exactly the dead sector named, and green $\geq$ red $>$ blue in both eyes. All twelve pass (Fig. 3): right eye 3.9 arcmin and 0.031 with sector six silent, left eye 7.8 arcmin and 0.067 with every sector responding, and the response phase within 42 to 47° in every responding sector of both eyes and random in the silent one. A first run of the gate had lost one contrast step to noise and fitted the left eye's contrast from the saturating top of its curve, four times too low; that is why the fit anchors on the step below the finest responding one, as the sweep method does by spanning the fall to the noise floor.

IV. First sitting

The first sitting on a person (one of the authors) ran the right eye with the left covered, then the left, and a second run with both eyes open, at 12 Hz on the Quest, in one evening. No threshold could be fitted. The only clear response was the full-patch colour flicker of the right eye, a 12 Hz line on Oz at a $\log_{10}$ signal-to-noise ratio of $+0.75$ in ON blocks against -0.22 in OFF blocks (Fig. 4b); the gratings gave a weak response at the coarsest bars (Fig. 4c), the sectors and contrast steps none. The raw EEG says why (Fig. 4a): both ear electrodes sat at 100 to 280 μV rms for the whole sitting, loose under the headset strap, while the signal check that opens every session had reported them good because it was written to catch an electrode floating at the amplifier rails and accepts anything below 350 μV; the Oz electrode was clean for the right eye (13 to 18 μV) and rose to 50 to 77 μV afterwards; and the patch was 0.6 of a browser window that was not full screen. The sitting changed the apparatus more than the phantom had: before each eye the exam now measures every channel's rms and, when any is above 60 μV, shows an electrode check on the headset and waits for a trigger pull until every channel holds; the scorer sets a channel out of contact aside; the read-

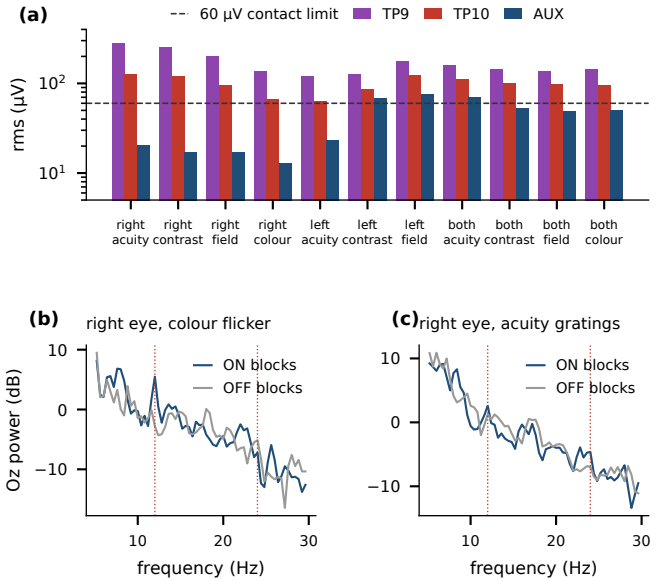


Fig. 4. The first sitting's raw EEG. (a) Rms of the two ear electrodes and of Oz through every test of the two runs against the 60 μV contact limit: the ears never held, Oz held for the right eye only. (b) Oz spectrum in the ON blocks against the OFF blocks of the right eye's colour test, with the 12 Hz line and its double marked: the one clear response of the sitting. (c) The same for the right eye's acuity test.

out names it and refuses to extrapolate a single responding step into a number; the page goes full screen on the exam's own trigger pull with a patch the full height of the window.

V. Discussion and Next Steps

What is established is the apparatus and its validation: four tests per eye that run without a response from the subject on a headband and a headset sold to consumers, a scorer with an exact test per step, a threshold rule taken from the sweep VEP, and a phantom gate that catches the failures a human sitting is too slow and too dear to find. What is not established is any measurement on a person; the first sitting is reported as the reason the contact check exists, not as a result. Two things are open before a second sitting can mean anything: the display's pixels per degree, which sets the arcmin scale and has not been measured on the headset (the comparison between the eyes does not need it, a Snellen value does), and the flicker rate for the pattern tests, since pattern onset at 12 Hz shows the grating for three frames at a time and a slower frame-exact rate such as 6 Hz may drive the response that the luminance flicker of the colour test already drives. After that the road is the usual one: a blur study with trial lenses against a letter chart, then a normative cohort. The SSVEP's sensitivity to an electrode that has moved a few millimetres is the practical limit of the consumer headband [4]; a screen built on it must check its own contact at every step, which this one now does.

References

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